

Eigenvalues and inverse problems

Unitary-orbit distance under finite block repetition

Difficulty: challenging **Importance:** interesting to specialist

Status: Partially resolved

Rating rationale: Challenging reflects preservation of a nontrivial spectral-norm orbit distance under larger unitary mixing; specialist impact concerns finite matrix amplification and operator-algebraic nearness.

Problem statement

For $A, B \in \mathbb{C}^{n \times n}$, define the distance between their unitary similarity orbits in the spectral norm by

$$\delta_n(A, B) = \min_{U \in \mathbb{C}^{n \times n}, U^*U = I_n} \|A - U^*BU\|_2.$$

For a positive integer k , write $A^{(k)} = I_k \otimes A$, the block diagonal matrix with k copies of A . Is it true that, for every $n \geq 3$, every finite integer $k \geq 2$, and every pair of normal matrices $A, B \in \mathbb{C}^{n \times n}$,

$$\delta_{nk}(A^{(k)}, B^{(k)}) = \delta_n(A, B)?$$

Normality means $AA^* = A^*A$ and $BB^* = B^*B$. The minimizing unitary on the left is unrestricted in dimension nk ; it may mix different copies. Thus the question asks whether block repetition can improve the best unitary-similarity fit between normal matrices.

Why it matters

This is a matrix-nearness problem with prescribed spectra. It tests whether a structured enlargement changes the optimum in spectral norm, a metric for which matching eigenvalues alone need not give the orbit distance.

References

- L. W. Marcoux and Y. Zhang, *On Specht's Theorem in UHF C-algebras**, Journal of Functional Analysis 280 (2021), 108778, §5, Question 5.4, pp. 26–27; [institutional full text](#).
- L. W. Marcoux, P. Sarkowicz, and Y. Zhang, *Kaplansky's problem and unitary orbits in matrix amplifications*, arXiv:2508.13834v1 (2025), §3.5 and Propositions 3.15, 3.21, and 4.14.

Earlier status check — 2026-09-08

Theorem 5.3 of the source proves equality for normal matrices of order two. Theorem 5.1 establishes a strict decrease for some unrestricted matrices, so normality cannot simply be omitted. The example after Question 5.4 uses infinitely many copies and does not resolve the finite- k question. The direct follow-up 2508.13834v1 records the order-two result after Corollary 3.5. The corollary itself proves equality for self-adjoint matrices of every order. Its normal-element Propositions 3.15 and 3.21 concern zero orbit distance; its Proposition 4.14 counterexamples have no normality requirement. Neither settles preservation of all distances for finite normal matrices. The published §5 and these later passages were checked. Searches for *Marcoux Zhang Question 5.4 normal matrices*, *normal matrices ampliations unitary orbit distance*, and 2025/2026 follow-ups found no finite-dimensional normal counterexample or general proof. This is limited search evidence, not a certificate of openness.

Audit update — 2026-09-10

Rechecked [Marcoux–Sarkowicz–Zhang](#), Corollary 3.5: equality is proved for every pair of self-adjoint matrices, in every dimension and finite multiplicity. This substantive normal subclass was missing from the earlier status explanation. General normal matrices remain unresolved in the checked sources; finite-normal amplification searches found no completion. The nonnormal counterexamples and zero-distance results do not settle the remaining positive-distance question.